



Course Description

COURSE NAME	Introduction to Information Retrieval
CODE	TECH 400
UNITS	3
LENGTH OF CLASS	8 weeks
COURSE DESCRIPTION	This course offers the fundamental principles, processes, and techniques essential for comprehending the theory, execution, and assessment of information retrieval (IR) systems. Its goal is to cultivate a comprehensive grasp of accessing information through information retrieval, covering foundational aspects of IR with a focus on the core of search engines, as well as delving into insights on diverse information types. By the end of the program, individuals will have acquired both theoretical knowledge and practical skills necessary for the successful development and assessment of robust information retrieval systems.
REQUIRED TEXT	Text Book Manning C., & Raghavan P., & Schutze H (2009), <i>An Introduction to Information Retrieval (Online ed.)</i>, Cambridge UP. (Freely available) http://www.informationretrieval.org/ Supplementary Books Hersh W (2020), <i>Information Retrieval: A Biomedical and Health Perspective (4th ed.)</i>, Springer ISBN: 978-3030476854 ISBN: 9783030476861 Goker A, & Davies J (2009), <i>Information Retrieval: Searching in the 21st Century (1st ed.)</i>, Wiley ISBN: 9780470027622
METHOD OF INSTRUCTION	The course is conducted in a combination modality. Students interact with each other and with the faculty in a classroom setting and in an online learning system. Learning will be facilitated through lecture-discussions, presentations, cooperative learning, and case studies.
SCOPE	Student outcomes are measured through professional individual assignments, discussion postings, comprehensive learning assessments, and class participation.



Summary of Graded Work and Assessments

Graded work and assessments offer students the opportunity to show the degree of mastery for each CLO. The following table shows how assessments and CLOs align (link).

Assignments	Totals	Weight	CLOs
Engagement and Professionalism - Laboratory Class Activities (Rubric)	160	16%	1, 2, 3, 4, 5
Week 1 Discussion	50	5%	1, 2
Week 2 Assignment	100	10%	1, 2, 5
Week 3 Assignment	100	10%	1, 2, 5
Week 4 Assignment	100	10%	2
Week 5 Discussion	50	5%	4
Week 5 Assignment	90	9%	2, 4
Week 6 Assignment	100	10%	1, 2, 4, 5
Week 7 Discussion	50	5%	4
Week 7 Assignment	100	10%	5
Week 8 Presentation	100	10%	1, 2, 3, 4, 5
Total Points/Percentage	1000 Points	100%	

Course Policies

For Westcliff's course policies, please see the [Course Policies](#) document.

Discussion Requirements

For all discussions, the primary response is due by Thursday at 11:59 p.m. Pacific Time. The primary response must be at least 200 words in length and fully address the topic, demonstrating critical thinking and understanding. Each student must then also post a minimum of two responses to other students in the discussion by Sunday night at 11:59 p.m. Pacific Time. Each peer response must be at least 50 words in length and substantively engage with the other student's original post, continuing the discussion in a professional manner. If at any time information or material is brought in from an outside source or website, it must be properly cited following APA 7th edition guidelines and a full reference must be provided.

Assignment Requirements

Each assignment deliverable is specifically defined in the assignment instructions, such as page length, citations and references, audio or video, presentations, tables, etc. For all written assignments, the required page length does not include the cover or references pages. Refer to the specific requirements as stated in each assignment, and reach out to your instructor for additional information as needed. All graded submissions are due by Sunday at 11:59 p.m. Pacific Time.

All written work must adhere to APA 7th edition academic formatting requirements including core components such as the cover page, page numbers, headings, citations, 1" margins, paragraph indentations, left alignment, double spacing throughout, and the final references using hanging indents

Participation Requirements

Students are required to attend each live class session either in person or virtually as stipulated in the course policies. Participation in the live class session is determined by actively engaging, answering or asking questions, providing comments, interacting in group activities, etc., as required by the instructor. Students who are unable to attend the live in-class or virtual sessions must follow the VCS submission requirements as stated in the Course Policies document.

Writing Center

The Westcliff University Writing Center is dedicated to providing quality support to students and faculty. From assignment review, to in-class workshops, to dissertation support, to publication help, the Writing Center is committed to empowering individuals to use the written language to articulate and disseminate knowledge.



Course Learning Outcomes (CLOs)

Learning outcomes are statements that describe significant and essential scholarship that students have achieved and can reliably demonstrate at the end of the course. Learning outcomes identify what the learner will know and be able to do by the end of a course – the essential and enduring knowledge, abilities (skills), and attitudes (values, dispositions) that constitute the integrated learning needed for successful completion of this course. The learning outcomes for this course summarize what students can expect to learn, and how this course is tied directly to the educational outcomes of the degree.

Course Learning Outcomes (CLOs)	PLOs
1. Implement document pre-processing and identification of indexes.	1, 3
2. Compare and contrast the various tools and technologies used for understanding search structures, information retrieval models, and query processing.	1, 3
3. Evaluate the information retrieval system.	1
4. Apply web crawler and methods from algebra and probability in web search.	1, 3
5. Solve and demonstrate real-world problems such as document ranking, search engines by the use of an information retrieval system.	1, 3



Detailed Course Outline

The following outline provides important assignment details for this course, unit by unit. Students are responsible for all of the assignments given. Please refer to the Detailed Description of Each Grading Criteria in the syllabus for specific information about each assignment.

Week 1

Assignments to complete this week:

- Reading: An Introduction to Information Retrieval
 - Chapter 1: Boolean Retrieval (p. 1 -16)
 - Chapter 2: The term vocabulary and postings lists (p. 19 - 44)
- Articles to Read
 - Geeksforgeeks: [Document Retrieval using Boolean Model and Vector Space Model](#)
 - Waller, W. G., & Kraft, D. H. (1979). [A mathematical model of a weighted Boolean retrieval system. *Information Processing & Management*, 15\(5\), 235-245.](#)
- Videos
 - Youtube Video: [NLU and Information Retrieval](#)
 - Youtube Video: [The Boolean Retrieval Model](#)

Week 1 Live Class Activities ([Rubric](#))

This week's live class activity involves an individual task to help students understand the basic boolean retrieval system implementation. In this activity, students need to perform the following tasks:

- Set up their environment.
- Successfully run an example of boolean retrieval.

The instructor will guide you in selecting tools and providing small boolean retrieval examples. Upon completion, students need to share their findings with the live class.

Week 1 Discussion ([Rubric](#))

Discuss the criticisms surrounding the Boolean retrieval technique and explore the reasons why, despite these critiques, it is still employed in certain situations. Discuss with a proper example.

Week 2

Assignments to complete this week:

- Reading: An Introduction to Information Retrieval
 - Chapter 3: Dictionaries and Tolerant Retrieval (p. 49 -64)
 - Chapter 4: Index Construction (p. 67 - 82)
- Videos
 - Youtube Video: [The Inverted Index](#)

Week 2 Live Class Activities ([Rubric](#))

The aim of this class activity is to understand and implement text preprocessing and indexing for information retrieval. Divide the class into groups based on the class size and each group should complete the following task:

- Text preprocessing, including parsing, stop word removal, stemming, etc.
- Indexing.

The instructor will assist in selecting samples for text preprocessing and indexing. After completing the tasks, students will have the opportunity to share and discuss their findings with the live class.

Week 2 Assignment ([Rubric](#))

In this assignment, students will create a basic Information Retrieval system for a small collection of 5-10 documents. The task includes writing a program to add documents to the system, making a basic dictionary, forming an inverted index for the document collection, and adding Boolean retrieval functions. The final submission should comprise a straightforward report explaining the choices made during design and implementation, providing an accessible hands-on experience with Information Retrieval concepts. Put the snapshot while writing and explaining the details of a report. Publish your code on [GitHub](#). Ensure to provide the [GitHub](#) link when submitting your assignment.



Week 3

Assignments to complete this week:

- Reading: An Introduction to Information Retrieval
 - Chapter 6: Scoring, term weighting and the vector space model (p. 109 - 131)
 - Chapter 7: Computing scores in a complete search system (p. 135 - 149)
- Readings:
 - Lee, D. L., Chuang, H., & Seamons, K. (1997). [Document ranking and the vector-space model](#). *IEEE software*, 14(2), 67-75.
 - Ramos, J. (2003, December). [Using tf-idf to determine word relevance in document queries](#). In Proceedings of the first instructional conference on machine learning (Vol. 242, No. 1, pp. 29-48).

Week 3 Live Class Activities ([Rubric](#))

The aim of this class activity is to comprehend and execute document ranking. Divide the class into groups according to its size. Each group are assigned the following tasks:

- Ranking documents based on user queries.

The instructor will provide guidance in selecting sample documents and the chosen method for implementation. After completing the task, students will have the chance to share and discuss their findings with the live class.

Week 3 Assignment ([Rubric](#))

For this task, gather 5-10 documents available online, and store them in .txt, .doc, or pdf formats.

The documents can include news articles or any publicly accessible content. Implement a similarity score calculation between the documents using a vector space model. Present your findings in a report, including snapshots to illustrate and explain the code and results. Publish your code on [GitHub](#). Ensure to provide the [GitHub](#) link when submitting your assignment.



Week 4

Assignments to complete this week:

- Reading: An Introduction to Information Retrieval
 - Chapter 8: Evaluation in information retrieval (p. 151 - 173)
 - Chapter 9: Relevance feedback and query expansion (p. 177 - 193)
- Online:
 - Bama, S. S., Ahmed, M. I., & Saravanan, A. (2015). [A survey on performance evaluation measures for information retrieval system](#). International Research Journal of Engineering and Technology, 2(2), 1015-1020.
 - Lavrenko, V., & Croft, W. B. (2017, August). [Relevance-based language models](#). In ACM SIGIR Forum (Vol. 51, No. 2, pp. 260-267). New York, NY, USA: ACM.
- Video:
 - Youtube Video: [Query expansion](#)

Week 4 Live Class Activities ([Rubric](#))

In this live class activity, students will understand and implement the evaluation system used in information retrieval. To accomplish this, divide the class into groups based on its size and complete the following tasks:

- Perform the evaluation of various methods in information retrieval, such as nDCG.

The instructor will assist in selecting evaluation methods and tools. After completing the task, students will have the opportunity to share and discuss their findings with the live class.

Week 4 Assignment ([Rubric](#))

In this assignment, students are tasked with producing a 2-3 page report adhering to APA format guidelines. The focus of the report is on an exploration of document ranking scoring functions, specifically Okapi BM25 and the Language model with Jelinek-Mercer Smoothing. The assignment requires a critical analysis of the practical application of these scoring functions, accompanied by suitable examples. Students should delve into the nuances of each scoring function, providing insightful commentary on their use and impact.

Week 5

Assignments to complete this week:

- Reading: Introduction to Information Retrieval
 - Chapter 11: Probabilistic Information Retrieval (p. 219 - 234)
 - Chapter 12: Language Models for Information Retrieval (p. 237 - 251)
- Video:
 - Youtube Video: [Probabilistic Retrieval Model Basic Idea](#)

Week 5 Discussion ([Rubric](#))

Discuss the differences between vector space relevance feedback and probabilistic relevance feedback. Engage with others in the class regarding their explanations and work toward a common understanding.

Week 5 Assignment ([Rubric](#))

Passage retrieval is a specialized type of IR application that retrieves relevant passages (or pieces of text) rather than an entire ranked set of documents. You can take the reference of the following article or other similar articles. Prepare a two-page research report by highlighting different methods used in passage retrieval with their strengths and weaknesses.

Reference: [A passage retrieval method based on probabilistic information retrieval model and UMLS concepts in biomedical question answering](#)

Week 6

Assignments to complete this week:

- Reading: An Introduction to Information Retrieval
 - Chapter 13: Text Classification and Naïve Bayes (p. 253 - 285)

Week 6 Live Class Activities ([Rubric](#))

The objective of this class activity is to utilize classification, specifically a machine learning model, for categorizing text, a common practice in information retrieval. Divide the class into groups based on its size and each group will complete the following tasks:

- Perform text classification using techniques such as Naive Bayes.

The instructor will assist in selecting the dataset and tools. Upon completing the task, students will have the chance to share and discuss their findings with the live class.

Week 6 Assignment ([Rubric](#))

Propose and create a basic information retrieval system of your choice using a publicly available dataset, such as [CISI \(a dataset for Information Retrieval\)](#) from Kaggle. Design the system architecture, including preprocessing steps, indexing, retrieval model selection, and query processing mechanisms. Students should select from the various methods covered during earlier lectures. Implement them using a programming language of their choice. The report should include information about the dataset, system design, and implementation details. Present your findings in a report, including snapshots to illustrate and explain the code and results. Publish your code on [GitHub](#). Ensure to provide the [GitHub](#) link when submitting your assignment.



Week 7

Assignments to complete this week:

- Reading: Introduction to Information Retrieval
 - Chapter 19: Web Search Basics (p. 421 - 441)
- Video:
 - Youtube Video: [Stanford Seminar - Beyond Web Search](#)

Week 7 Discussion ([Rubric](#))

Two web search engines A and B each generate a large number of pages uniformly at random from their indexes. 30% of A's pages are present in B's index, while 50% of B's pages are present in A's index. What is the number of pages in A's index relative to B's? Discuss your answer with appropriate reasoning.

Week 7 Assignment ([Rubric](#))

In this assignment, students need to evaluate the performance of the models using different metrics such as recall, precision, MAP, nDCG etc for the chosen project (proposed in assignment 6). Prepare a report which contains the implementation details, codes, inline comments, screenshots of output and detail analysis. Present your findings in a report, including snapshots to illustrate and explain the code and results. Publish your code on [GitHub](#). Ensure to provide the [GitHub](#) link when submitting your assignment.

Week 8

Assignments to complete this week:

- Review all the chapters from week 1 to 7.

Week 8 Assignment Presentation ([Rubric](#))

Create a refined PowerPoint presentation offering a succinct summary of project findings (proposed in assignment week 6 & 7). The presentation should consist of at least the following information; your problem statement, dataset information, methodology used, evaluations, major findings, critical analysis, sample output and recommendations.